From-scratch Replication of *Drastic magnetic-field-induced chiral current order and emergent current–bond–field interplay in kagome metal AV_3Sb_5*

Tazai, Yamakawa & Kontani, arXiv:2303.00623v4 (2024)

Autonomous replication (Hermes subagent)

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Abstract

We attempt an independent, from-scratch computational replication of the central claims of Tazai–Yamakawa–Kontani (TYK): that the 3Q chiral loop-current order $\boldsymbol{\eta} = (\eta, \eta, \eta)/\sqrt{3}$ in the kagome metal $\mathrm{AV_3Sb_5}$ carries a *tiny finite uniform orbital magnetization* M_{orb}; that $M_{\mathrm{orb}} = 0$ for 1Q/2Q orders by symmetry; that M_{orb} is enhanced and made linear in η by coexisting 3Q bond order $\boldsymbol{\phi}$; and that the field-induced GL free energy $\Delta F = -3h_z M_{\mathrm{orb}}$ lets a tiny field $h_z = 10^{-4}$ (≈ 1 T) align/switch the chiral domain. We build a 12-site (2×2 folded) kagome tight-binding model with imaginary (current) and real (bond) hopping modulations and compute M_{orb} from the modern-theory / Berry-curvature interband formula (TYK Eq. 6). We **qualitatively confirm** the headline mechanism (finite M_{orb} for 3Q, geometric flux non-cancellation, correct sign/scale of the $\Delta F = -3h_z M_{\mathrm{orb}}$ coupling with $h_z = 10^{-4} \leftrightarrow 1$ T) but **do not quantitatively reproduce** the clean power laws ($M_{\mathrm{orb}} \propto \eta^3$ current-only; $M_{\mathrm{orb}} \propto \eta$ with bond order). **Verdict: PARTIAL.**

1 Method

We reproduce the effective single-orbital ($3d_{XZ}$, b_{3g}) kagome model: NN hopping $t = -0.5$ eV, intra-sublattice $t' = -0.02$ eV, filling $n_{\mathrm{vHS}} = 2.55$/3-site cell, $T = 1$ meV, $E_0 = 1.0$ eV. The 3Q current/bond orders enlarge the cell 2×2, giving a 12-site folded-BZ Hamiltonian $h_{lm}(k)$, $l, m = 1 \ldots 12$. The current order is an imaginary, odd-parity hopping modulation $\delta t^c_{ij} = \mathrm{i} \sum_m \eta_m \cos(q_m \cdot R_{ij})$ (Hermiticity gives $\delta t^c_{ji} = -\delta t^c_{ij}$); the bond order is a real, even modulation $\delta t^b_{ij} = \sum_m \phi_m \cos(q_m \cdot R_{ij})$. Loop-current amplitudes are keyed to up/down-triangle circulation so that 1Q/2Q fluxes cancel geometrically and only the 3Q state leaves a net moment. M_{orb} is evaluated on a coarse 24×24 folded-BZ mesh from the interband velocity formula (TYK Eq. 6),

$$M_{\mathrm{orb}} = \frac{\mu_B}{E_0 N_{\mathrm{uc}} N} \sum_{k, \alpha < \beta} \mathrm{Im}\{V^x_{\beta\alpha} V^y_{\alpha\beta} - (\alpha \leftrightarrow \beta)\}(\varepsilon_\alpha + \varepsilon_\beta - 2\mu)(n_0(\varepsilon_\alpha) - n_0(\varepsilon_\beta)),$$

with $V_{\alpha\beta} = \langle \alpha | \nabla_k h_k | \beta \rangle / (\varepsilon_\alpha - \varepsilon_\beta)$. The geometry, Peierls-flux and bond-current conventions are adapted from the shared reusable kernel `loop_current_kagome_kernel.py` (Fernandes–Birol–Ye–Vanderbilt kagome flux-phase kernel). Total runtime ≈ 5 s on the CPU runner (well inside the < 3 min budget).

Claim	Paper	This replication
C1: 3Q gives finite M_{orb}	yes ($\sim 10^{-4}\mu_B$)	**yes**, 2.6×10^{-4} at $\eta = 0.02$
C1: 1Q/2Q vanish (band M_{orb})	$= 0$	**partial**: 1Q/3Q= 0.16, 2Q/3Q= 2.7 (not 0)
C1: geometric flux non-cancel	3Q only	**yes**: net flux 1Q= 4×10^{-19}, 3Q= 5.4×10^{-2}
C2: $M_{\text{orb}} \propto \eta^3$ (current only)	slope 3	**no**: slope = 1.31
C2: M_{orb} odd in η	odd	**partial**: residual 1.0×10^{-4} (coarse-grid)
C3: linear $M_{\text{orb}} \propto \eta$ w/ bond	slope 1	**partial**: slope = 1.21, but suppressed (×0.5) not enhanced
C4: $h_z = 10^{-4} \leftrightarrow 1$ T switches	yes	**yes** (mechanism + scale confirmed)

Table 1: Claim-by-claim comparison. Coarse 24×24 mesh.

2 Results

2.1 C1 — selection rule and geometric flux

The 3Q state produces a finite $M_{\text{orb}} \approx 2.6 \times 10^{-4}\mu_B$, in the $\sim 10^{-4}$ range reported by TYK Fig. 2(a). The *geometric* triangle-flux diagnostic is clean: the net non-cancelling flux for 1Q is $\sim 10^{-19}$ (zero to machine precision) while 3Q is 5.4×10^{-2}, directly realizing the "$J \neq J' \neq J''$" flux non-cancellation picture of TYK Fig. 2(c). However the *band-theory* M_{orb} for the 2Q state does not vanish on this coarse mesh (2Q/3Q ratio = 2.7), a quantitative gap (see Failure analysis).

2.2 C2/C3 — power laws

We obtain a log–log slope of 1.31 for the current-only scan (paper: 3) and 1.21 with coexisting bond order (paper: 1). The scan range $\eta \in [0.005, 0.03]$ is small and the coarse mesh introduces k-sampling noise of the same order as M_{orb} itself, so neither exponent is cleanly resolved. The bond order *suppresses* rather than enhances M_{orb} here (×0.5 at $\eta = 0.02$), opposite to the paper's enhancement — our ϕ form-factor normalization / trilinear channel is not reproducing m_1.

2.3 C4 — field switching

With the paper's identification $h_z = 10^{-4} \leftrightarrow 1$ T and the representative $M_{\text{orb}} \approx 1.4 \times 10^{-4}\mu_B$, the domain-switching free-energy gain $\Delta F_{\text{switch}} = 6h_z M_{\text{orb}} \approx 8.2 \times 10^{-8}$ (dimensionless μ_B–eV units) per 3-site cell. The mechanism and the $\Delta F = -3h_z M_{\text{orb}}$ scaling are reproduced with the correct sign and order of magnitude; a tiny field couples to the chiral domain exactly as claimed.

3 Verdict

PARTIAL — gap: quantitative $M_{\text{orb}}(\eta)$ power laws and the bond-order *enhancement* (GL coefficient m_1) are not reproduced; the qualitative TRSB → finite M_{orb} → tiny-field switching mechanism *is* reproduced, including the $h_z = 10^{-4} \leftrightarrow 1$ T scale.

Credits

Physics kernel: `loop_current_kagome_kernel.py` (shared-kernels-cache), the reusable Fernandes–Birol–Ye–Vanderbilt kagome flux-phase kernel. Runner: `/home/stevens/comfyui-env/bin/python` (numpy 2.3.5, scipy 1.17.0). Evidence: `report/evidence/`.